

Introduction to Survival Analysis

The DARTH Workgroup

What is survival analysis?

- ▶ Set of statistical methods that can handle:
 - ▶ Data skewness
 - ▶ Censoring
 - ▶ (Time-dependent) covariates
 - ▶ State transition processes
 - ▶ Recurrent events
 - ▶ The expectation of cure

Key Concepts in survival analysis

- ▶ survival to time T : the random variable capturing time since the beginning of the event.
- ▶ Survival: the probability of not experiencing an event unit some time t $S(t) = Pr(T > t) = 1 - Pr(T \leq t)$
- ▶ Cumulative density function:

$$F(t) = Pr(T \leq t)$$

$$F(t) = 1 - S(t)$$

- ▶ Hazard function (the instantaneous rate of the event):

$$h(t) = \lim_{\delta t \rightarrow 0} \frac{1}{\delta t} \frac{F(t+\delta t) - F(t)}{S(t)}$$

turns out $h(t) = \frac{f(t)}{S(t)}$

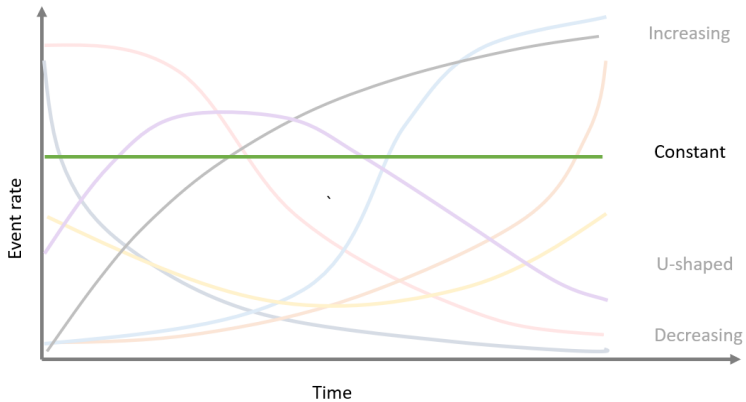
- ▶ Cumulative hazard $H(t) \int_0^t h(x) dx$

as well as $S(t) = e^{-H(t)}$

Example: the exponential distribution

- ▶ $H(t) = \lambda t$
- ▶ $S(t) = \exp(-\lambda t)$
- ▶ $f(t) = \lambda \exp(-\lambda t)$
- ▶ Weibull , gamma, lognormal etc. . .

Some shapes of hazards



Types of survival analysis models

- ▶ Non-parametric models (e.g. Kaplan-Meier)
- ▶ Semi-Parametric models (e.g. Cox Proportional Hazard)
- ▶ Parametric models (e.g. Accelerated Failure Time)

Fully Parametric models

- ▶ Model either expected time or hazard rate
- ▶ Resemble a regression model but can capture censoring
- ▶ Fit a distribution to the (observed) time-to-event data
 - ▶ 1 parameter : exponential
 - ▶ 2 parameter : Weibull, lognormal, gamma etc
 - ▶ 3, 4 parameter: Gen. gamma , Gen F etc
- ▶ Assumption that censored patients will follow similar patterns to the observed.
- ▶ Temporal extrapolation possible!

Choosing the “right” distribution

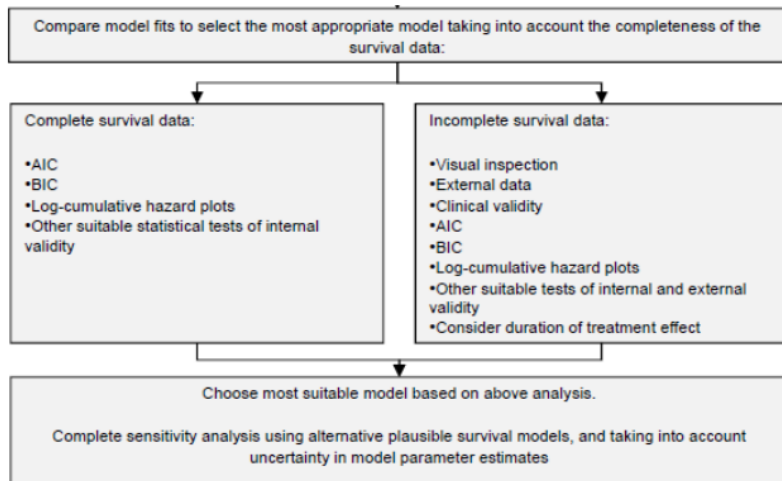
- ▶ Statistical considerations of goodness-of-fit
- ▶ Clinical plausibility
 - ▶ in-sample
 - ▶ extrapolations
 - ▶ out-of-sample

Goodness of fit metrics

- ▶ Likelihood based metrics:
- ▶ Akaike Information Criterion (AIC)
- ▶ Bayesian Information Criterion (BIC)

Choosing the “right” distribution

Latimer et al 2011 (DSU 14)



Approaches in extrapolation

Modeler faced with more decisions:

- ▶ Extrapolate using KM and “fitted” tails
- ▶ Extrapolate using the fitted curve
- ▶ Fit separate curves to the data
- ▶ Extrapolate treatment effect as relative
- ▶ What is the behaviour of the relative effect post trial?

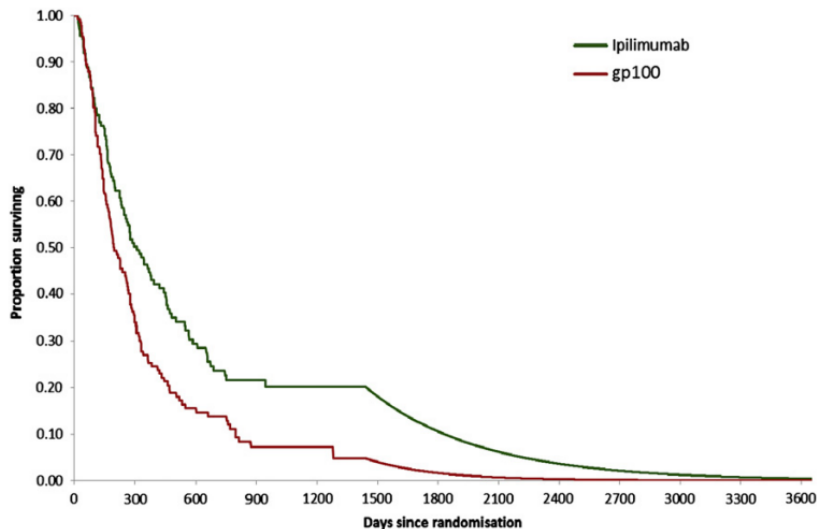


The Ends Justify the Mean: Outcome Measures for Estimating the Value of New Cancer Therapies

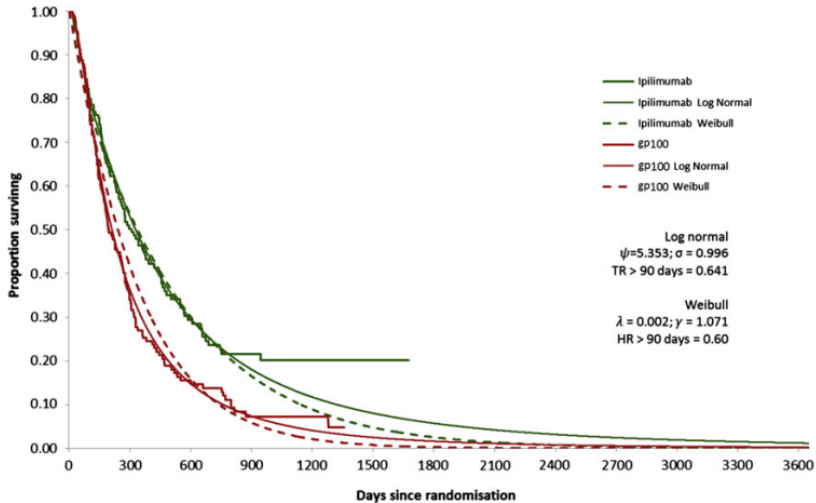
Andrew Davies, MSc^a, Andrew Briggs, DPhil^b, John Schneider, PhD^b, Adrian Levy, PhD^c, Omar Ebeid, MPH^b, Samuel Wagner, PhD^d, Srividya Kotapati, PharmD^d, Scott Ramsey, MD, PhD^e

Approaches in extrapolation

- ▶ Extrapolate using KM and “fitted” tails



► Extrapolate using the fitted curve



Extrapolating a relative treatment effect

Drummond, et al (2015). Methods for the economic evaluation of health care programmes. Oxford university press.

Should the hazard ratio change after the trial period?

